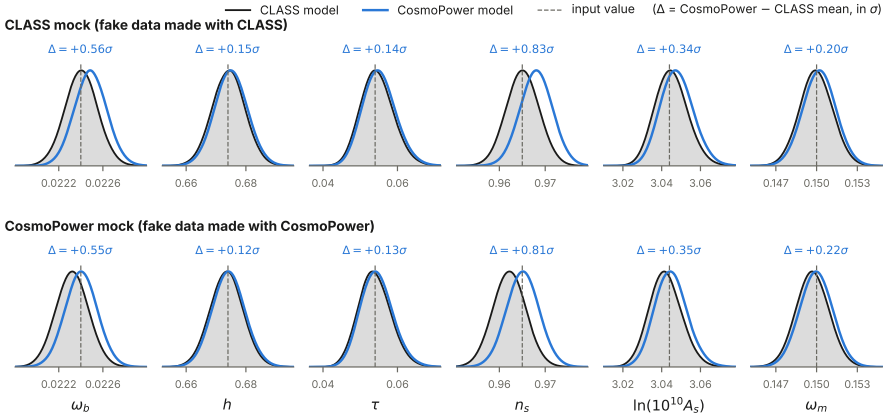
How much does $\partial C_\ell/\partial \omega_m$ depend on the other parameters?

At the benchmark-1 point, $\partial C_\ell/\partial \omega_m$ recomputed with each ΛCDM parameter θ_i shifted by ±3%; change measured with Planck-like noise weighting (33 μK-arcmin, 7′ beam, $f_{\rm sky} = 0.57$).

	change in $\partial C_\ell/\partial \omega_m$, θ_i shifted ±3%				same change at a
shifted θ_i	TT	TE	EE	$\phi\phi$	1σ shift (TT/TE/EE)
h	10.5%	11.0%	8.9%	1.7%	$\sim 3\%$
ω_b	4.6%	4.5%	3.4%	1.3%	$\sim 1\%$
A_s	2.8%	2.7%	2.8%	3.0%	$\sim 0.8\%$
n_s	2.0%	1.6%	1.7%	5.3%	$\sim 0.3\%$
τ	0.6%	0.8%	0.9%	0.1%	$\sim 1.6\%$

For a 1σ shift in any other parameter, $\partial C_\ell/\partial \omega_m$ changes by at most $\sim 3\%$ (from h). A ±3% step is a much larger move than 1σ for h, ω_b, A_s, n_s, and a smaller one for τ, hence its larger 1σ value.

Benchmark: CLASS vs CosmoPower, each fitted to both mocks



ΛCDM fit (one ω_m bin) to two mocks of the same cosmology, made with CLASS and with CosmoPower. 8 chains $\times$ 10^5 steps per run.

Each code recovers its own mock ($|\text{pull}| \leq 0.12\sigma$). On either mock CosmoPower sits above CLASS by the same amount: n_s $+0.8\sigma$, ω_b $+0.55\sigma$, A_s $+0.35\sigma$, ω_m $+0.2\sigma$; h, τ $\leq 0.15\sigma$. Errors identical.

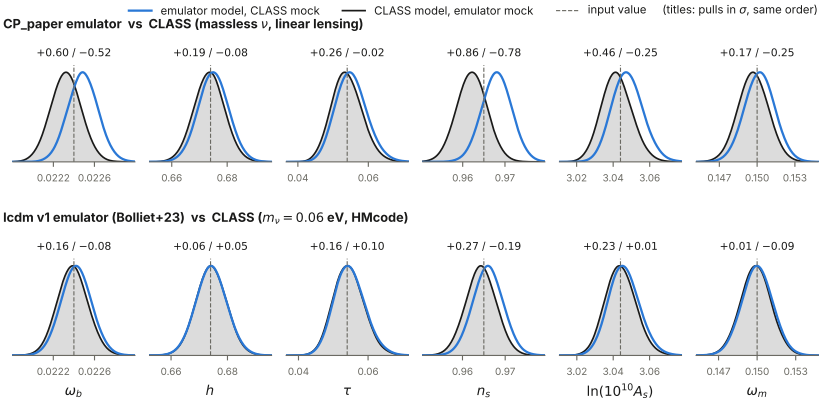
Benchmark: numbers, and comparison with the Fisher forecast

	CosmoPower − CLASS mean [σ]			error ratio	pull, code on its own mock [σ]	
	CLASS mock	CP mock	Fisher forecast	CP/CLASS	CLASS	CosmoPower
ω_b	+0.56	+0.55	+0.55	0.98	+0.03	+0.03
h	+0.15	+0.12	+0.13	1.00	+0.04	+0.04
τ	+0.14	+0.13	+0.15	1.01	+0.12	+0.11
n_s	+0.83	+0.81	+0.83	0.99	+0.03	+0.03
$\ln(10^{10}A_s)$	+0.34	+0.35	+0.37	1.00	+0.12	+0.10
ω_m	+0.20	+0.22	+0.21	1.00	−0.03	−0.03

- The shift does not depend on which code made the mock (agreement to $\leq 0.03\sigma$): it is a fixed offset between the two models.
- The linear Fisher forecast (CosmoPower model on a CLASS mock, deck 1) predicted every shift to within 0.03σ.
- Self-fits are unbiased, so the MCMC machinery is fine; the offset is the emulator's accuracy at the damping tail.

Pull = (mean − input)/σ. Numbers in `data/bench_mocks.json`, made by `scripts/fig_bench_mocks.py`.

Benchmark with the newer emulator: smaller code bias



Cross fits only: each code fitted to the other code's mock (ΛCDM, one bin, same input cosmology). Top: CP_paper networks (Spurio Mancini et al. 2022) vs CLASS with massless ν. Bottom: CosmoPower `lcdm v1` networks, Bolliet et al. 2023 (arXiv:2303.01591), vs CLASS with the same physics (1×0.06 eV ν, HMcode lensing).

Largest pull drops from 0.86σ (n_s) to 0.27σ; ω_b $0.60 \to 0.16\sigma$, A_s $0.46 \to 0.23\sigma$, ω_m $0.25 \to 0.09\sigma$. Errors unchanged.

The SPT emulator: what it is and how close to CLASS

CosmoPower `SPT_high_accuracy` ΛCDM networks, Balkenhol et al. (SPT-3G 2018 TT/TE/EE), trained on high-precision CAMB.

- Inputs $\omega_b, \omega_c, h, \tau, n_s, \ln(10^{10} A_s)$; lensed TT, TE, EE for $\ell = 2$–5000, in $\mu\mathrm{K}^2$. **No lensing ($\phi\phi$) network.**
- Training box (inferred): $\omega_c \in [0.090, 0.160]$, $h \in [0.55, 0.80]$, $\tau \leq 0.090 \Rightarrow$ each ω_m bin must stay in $\sim [0.112, 0.182]$.
- Physics: one massive ν (0.06 eV) and HMcode lensing, as `lcdm v1`.

	CP_paper	`lcdm v1` (Bolliet+23)	`SPT_high_accuracy`
trained on	CAMB, massless ν	CLASS, m_ν, HMcode	CAMB, m_ν, HMcode
C_ℓ vs matching CLASS ($\ell \leq 2508$)	TT -0.7%, EE -1% (tail)	$\sim 0.1\%$ TT, EE	0.06% TT, 0.08% EE rms
ℓ range	2–2508	2–11 000	2–5000
$\phi\phi$	yes	yes	no
parameter bias	$\leq 0.86\sigma$ (MCMC)	$\leq 0.27\sigma$ (MCMC)	$\leq 0.17\sigma$ (Fisher)

Bias: cross fits of the emulator model to a CLASS mock (TT, TE, EE + lensing, Planck-like noise). SPT: Fisher forecast, TT/TE/EE from SPT; with `lcdm v1`'s $\phi\phi$ added the forecast bias is $\leq 0.2\sigma$.

Estimating $\partial C_\ell/\partial\omega_m$ at fixed θ_s

CLASS, **numerical** (benchmark): five-point stencil, $\epsilon = 1.2\times10^{-3}$,

$$\frac{\partial C_\ell}{\partial \omega_m} \simeq \frac{C_\ell(-2\epsilon) - 8C_\ell(-\epsilon) + 8C_\ell(\epsilon) - C_\ell(2\epsilon)}{12\,\epsilon},$$

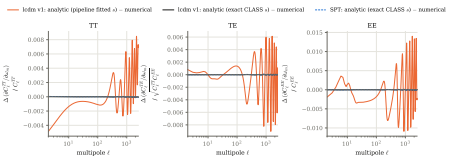
with h re-solved at every point so that CLASS's $100\theta_s = 1.0419693546$ (achieved to 4×10^{-9}). High precision: $\ell_{\max} = 3500$, refined ℓ and q sampling, tight tolerances (∼9 min/run).

Emulators, **analytic**: exact network Jacobian (no step), along the fixed-θ_s direction

$$\left.\frac{\partial C_\ell}{\partial \omega_m}\right|_{\theta_s} = \left.\frac{\partial C_\ell}{\partial \omega_c}\right|_{h} + \left.\frac{\partial C_\ell}{\partial h}\right|_{\omega_c} \left.\frac{dh}{d\omega_m}\right|_{\theta_s}, \qquad \left.\frac{dh}{d\omega_m}\right|_{\theta_s} = -3.524$$

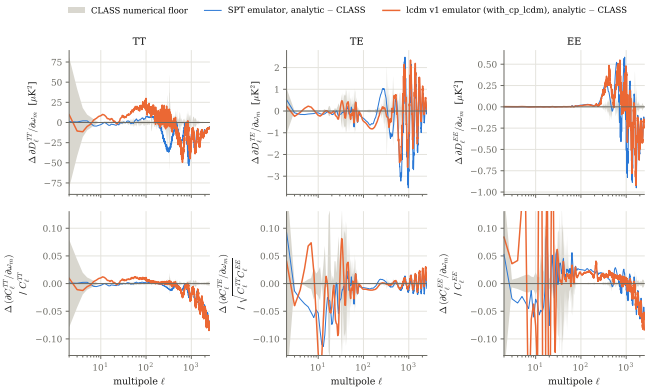
(slope from CLASS; $d\ln h/d\ln\omega_m = -0.748$).

Fiducial: ω_b 0.02237, ω_c 0.12, h 0.674, τ 0.054, n_s 0.965, $\ln(10^{10}A_s)$ 3.044, m_ν 0.06 eV, HMcode.



Analytic − numerical, same emulator, per unit ω_m, /C_ℓ. With the exact slope: $\lesssim 10^{-4}$ (Jacobians exact). The pipeline's fitted slope (−0.74844 vs −0.74773) alone moves the `lcdm v1` derivative by up to 0.8% (TT), 1.4% (EE).

$\partial C_\ell / \partial \omega_m$: emulators vs CLASS



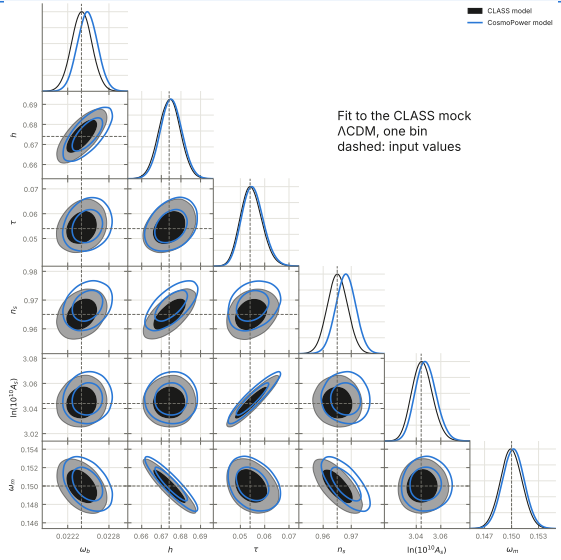
Top: absolute (μK^2 per unit ω_m). Bottom: per unit ω_m, relative to C_ℓ (TE: $\sqrt{C^{TT} C^{EE}}$). Grey: CLASS numerical floor.

$\sqrt{\chi^2}$, 1σ shift	TT	TE	EE
template itself	2.04	1.92	3.04
SPT − CLASS	0.071	0.026	0.063
lcdm v1 − CLASS	0.080	0.030	0.067
CLASS floor	0.005	0.006	0.009

$\delta\omega_m = 0.0012$; cosmic variance only, $f_{\rm sky} = 0.57$, $30 \le \ell \le 2508$.

Both emulators reproduce the CLASS derivative to 1.5–4%, SPT slightly better. The error grows with ℓ with the same shape for both: TT −4% of C_ℓ per unit ω_m at $\ell = 1500$–2000, −7 to −8% at 2500.

Backup: full posteriors, CLASS mock



Backup: full posteriors, CosmoPower mock

